Status	Finished
Started	Thursday, 1 October 2026, 4:10 PM
Completed	Thursday, 1 October 2026, 4:16 PM
Duration	5 mins 55 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search

b. Binary search ✓

c. Linear search

d. Sequential search

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. There should be direct access to the middle element in any sublist

c. There must be a mechanism to delete and/or insert elements in the list ✓

d. The list must be sorted

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list

b. In the right sublist

c. In the left sublist ✓

d. In the entire list

Question **4**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Binary search
b. Jump search
c. Linear search ✓
d. Hashing

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search requires the list to be sorted.

Question **6**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)
b. O(1)
c. O🥊 ✓
d. O(log n)

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By using a hash function
c. By dividing the list length by two ✓
d. By comparing each element sequentially

Question **8**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. The search is considered unsuccessful ✓
c. An error is raised
d. The first element is returned

Question **9**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94
b. 90 and 100
c. 90 and 99 ✓
d. 94 and 99

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements
b. When the list is unsorted
c. When the list is sorted ✓
d. When the list is very small

Question **11**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(1) ✓
b. O(n log n)
c. O(log n)
d. O👎

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
c. There must be a mechanism to access middle element directly
d. Must use a sorted array

Question **13**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion

b. Bubble ✓

c. Complexity

d. Selection

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94

b. 90 and 99 ✓

c. 90 and 94

d. 89 and 99

Question **15**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O🦴

b. O(1)

c. O(n log n)

d. O(log n) ✓